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Prof. Dr. Cheng-Jian Xu
Clinical Bioinformatics Group
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Re: PhD Position (f/d/m) in Clinical Bioinformatics

Dear Prof. Xu,

I am applying for the PhD position in Clinical Bioinformatics in your group. The project's core problem — reading immune aging, multimorbidity, and disease susceptibility off integrated multi-omics and population data, and turning that reading into predictive models for stratification — is the exact problem I have been working on independently for the past year, both as computational infrastructure and as a formal method for representing disease state across biological scales. I want to describe that work concretely, because the fit is specific rather than general.

The central object in my work is a *disease-state vector*. In my paper *Syndrome: Categorical Resolution of Disease Trajectories*, a patient's pathological signature is encoded as a vector $\mathbf{D} \in [0, 1]^8$ over eight cellular oscillator classes — protein folding, enzymatic catalysis, channel gating, membrane potential, ATP synthesis, genetic expression, calcium signalling, and circadian rhythm — each contributing a coherence index $\eta_i \in [0, 1]$. This is directly a multimorbidity representation: distinct conditions occupy distinct regions of the same low-dimensional space, so comorbidity, trajectory, and patient stratification become geometric questions on a shared manifold rather than separate disease labels. Optimal intervention targets are selected as $\arg \max_i w_i D_i$, a sparse ℓ_1 linear programme with a computable a priori side-effect bound, and disease trajectories resolve in $\mathcal{O}(N \log N)$. This is the kind of predictive, stratifying model your project aims to build, and it is designed to take its inputs from exactly the data types you work with.

A companion paper, *On Disease Epistemology in Blind Receiver*, addresses the diagnostic criterion itself. It proves that template-based diagnosis — comparison against a healthy reference — is epistemologically insufficient, and derives the unique viable alternative: loop-holonomy self-consistency of the biochemical circuit. A system is healthy iff the composed constraint map around every reaction cycle returns the identity ($H_\ell = \text{Id} \forall \ell$); disease is $H_\ell \neq \text{Id}$ for some cycle. The holonomy test reaches $\text{AUC} = 1.00$ against 0.896 for template comparison across 25 validation experiments. The relevance to immune aging is direct: aging is not a single-marker event but a gradual loss of circuit-level consistency that individual biomarkers miss precisely because it is distributed across the network. A criterion that detects distributed inconsistency is the right instrument for a process that is itself distributed.

On the immune side specifically, my paper *On the Thermodynamic Consequences of Categorical Completion on Biological Systems* models adaptive immunity as categorical filtering rather than molecular complementarity, and yields an empirical result on real immunopeptidomics data: MHC binding affinity correlates with parent-protein categorical richness independently of classical sequence features ($r = 0.73$, $p < 10^{-12}$, $n = 2,847$ IEDB peptides for MHC-I), and adding this term to NetMHCpan improves AUC from 0.68 to 0.81 ($p < 0.001$, DeLong test). This is the immune-function layer your project studies, treated with the same modelling discipline.

The computational infrastructure behind this is built and validated against authoritative reference data, and it maps onto the omics layers named in your posting. [gospel](#) integrates whole-genome VCF data with RNA-seq expression matrices at 10^6 variants/second, validated against 1000 Genomes Phase 3, gnomAD v3.1.2, and UK Biobank — this is genomic and transcriptomic integration at population scale, the core technical requirement of the role. [lavoisier](#) is a mass-spectrometry metabolomics pipeline achieving 98.9% feature-extraction accuracy against NIST reference libraries, with distributed computing via Ray and PyTorch models — the metabolomic omics layer. Both are deployable and documented, not one-off scripts; I build reproducible, reference-validated tools, and several are available in the browser without installation ([syndrome-seven.vercel.app](#)).

The disease-vector formulation extends naturally to your epigenetic focus: the epigenome enters as an additional coherence layer, and a methylation-derived biological-aging coordinate sits directly alongside the transcriptomic and genomic ones on the same manifold. The statistical and integration machinery I have built is layer-agnostic by design, so adding an omics modality is a question of ingestion and preprocessing rather than of reworking the model.

My academic background is in computational lipidomics at the Technische Universität München, where I trained in the multi-omics statistical methods that underpin all of this work; R and Python are both standard tools from that training. I am fully authorised to work in Germany without restriction or sponsorship (permanent residence), and I am available for the 1 September 2026 start. I am not proposing to import my framework wholesale into your programme — that is not the purpose of a doctoral position. I am proposing that a candidate who has independently derived and empirically validated a low-dimensional model of disease state, and built the multi-omics integration infrastructure your project depends on, will contribute to your work on immune aging and stratification from early on. I would welcome the chance to discuss it with you directly.

Yours sincerely,

Kundai Farai Sachikonye